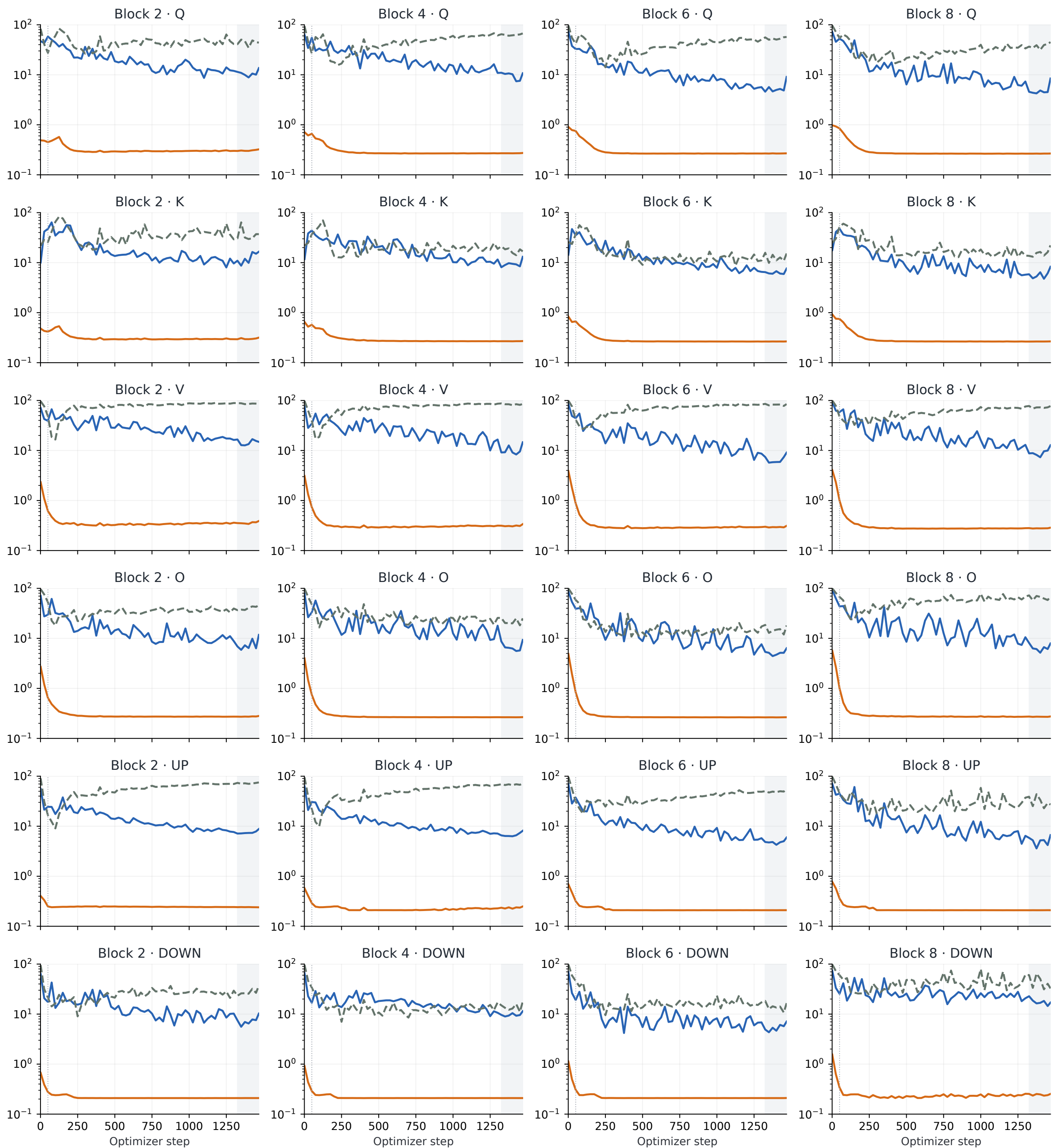


Energy in the leading singular direction · all 24 matrices

AdamW adaptive update Muon post-NS update Muon pre-NS momentum



$100 \times \sigma_1^2 / ||\text{matrix}||^2_F$. Lower values mean less concentration in one direction, not necessarily better learning.